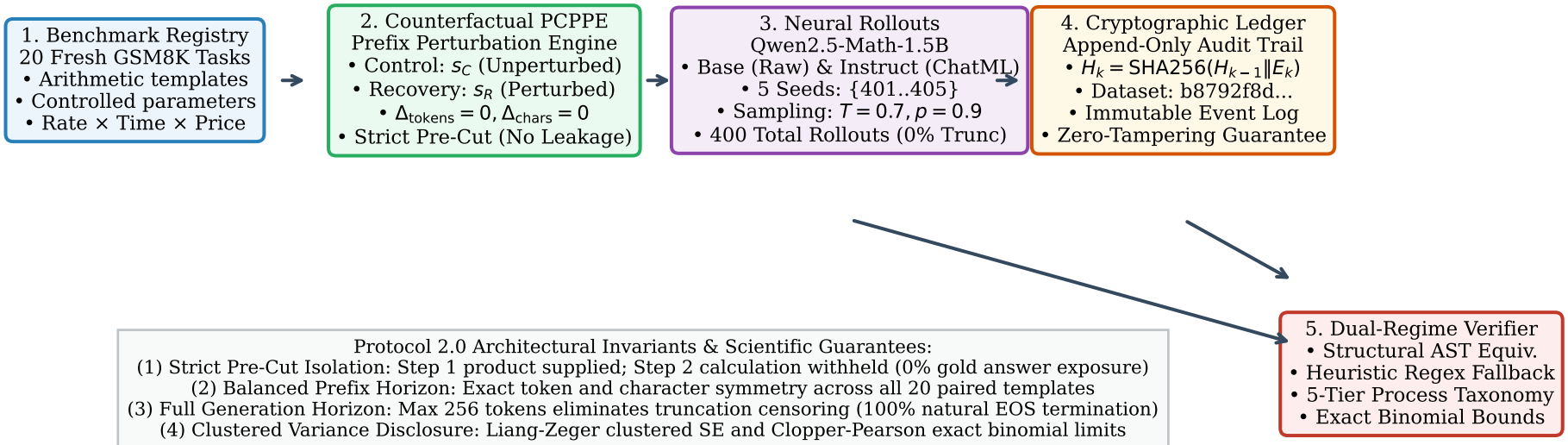
1. Benchmark Registry
20 Fresh GSM8K Tasks
• Arithmetic templates
• Controlled parameters
• Rate × Time × Price
2. Counterfactual PCPPE
Prefix Perturbation Engine
• Control: s_C (Unperturbed)
• Recovery: s_R (Perturbed)
• $\Delta_{\text{tokens}} = 0, \Delta_{\text{chars}} = 0$
• Strict Pre-Cut (No Leakage)
3. Neural Rollouts
Qwen2.5-Math-1.5B
• Base (Raw) & Instruct (ChatML)
• 5 Seeds: {401..405}
• Sampling: $T = 0.7, p = 0.9$
• 400 Total Rollouts (0% Trunc)
4. Cryptographic Ledger
Append-Only Audit Trail
• $H_k = \text{SHA256}(H_{k-1} \| E_k)$
• Dataset: b8792f8d...
• Immutable Event Log
• Zero-Tampering Guarantee
5. Dual-Regime Verifier
• Structural AST Equiv.
• Heuristic Regex Fallback
• 5-Tier Process Taxonomy
• Exact Binomial Bounds
Protocol 2.0 Architectural Invariants & Scientific Guarantees:
(1) Strict Pre-Cut Isolation: Step 1 product supplied; Step 2 calculation withheld (0% gold answer exposure)
(2) Balanced Prefix Horizon: Exact token and character symmetry across all 20 paired templates
(3) Full Generation Horizon: Max 256 tokens eliminates truncation censoring (100% natural EOS termination)
(4) Clustered Variance Disclosure: Liang-Zeger clustered SE and Clopper-Pearson exact binomial limits